

Bridging the Semantic Gap: Hierarchical Mid-level Interaction Graph Reasoning for Video Anomaly Recognition

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Abstract—Current CNN-based video models often struggle to connect low-level frame evidence with high-level behavioral semantics, leading to the loss of critical local details or global temporal context. To address this issue, we propose S-TRGR, a Spatio-Temporal Relation Graph Reasoning framework that explicitly targets mid-level semantic modeling. Our architecture combines a 2D CNN backbone for static feature extraction, a dual-Transformer design to capture spatial and temporal interactions, and a Graph Convolutional Network (GCN) for higher-order relational reasoning. By organizing video content into structured interaction graphs, S-TRGR encodes contextual cues that are essential for anomaly perception. Experiments on UCF-Crime and XD-Violence show that S-TRGR outperforms state-of-the-art methods, delivering a 5.0% absolute gain in Top-1 accuracy on UCF-Crime.

I. INTRODUCTION

Robust video understanding is fundamental to human-centric intelligence for shaping the digital future, yet it remains challenging because it requires jointly modeling spatial structure and temporal dynamics. Existing deep architectures often degrade on long-horizon, unconstrained surveillance videos, which in turn limits reliable abnormal behavior recognition [1]. To move beyond binary anomaly alarms and provide richer semantics, recent work increasingly adopts multi-class formulations [1]–[5]. However, long temporal spans, substantial environmental heterogeneity, and sparse fine-grained supervision still make it difficult to consistently associate entities and their interactions across diverse contexts.

CNN-based approaches typically rely on 3D convolutions (e.g., C3D [6], I3D [7], R(2+1)D [8], SlowFast [9], X3D [10]) to learn spatio-temporal representations. However, their ability to model global context is limited unless network depth or dilation is substantially increased, which becomes impractical for long-horizon videos. Transformer-based models (e.g., TimeSformer [11], ViViT [12]) capture long-range dependencies more naturally, but aggressive tokenization can attenuate fine-grained, high-frequency details and may overfit small datasets without strong regularization. Meanwhile, multimodal vision–language models (e.g., Vad-CLIP [3]) have shown strong performance on general tasks,

but their effectiveness on anomaly benchmarks such as UCF-Crime remains unclear.

Recent work has also explored weakly supervised frameworks for long-video analysis (e.g., [4], [13], [14]). However, the temporal coherence of anomalous events is often not sufficiently modeled, and MIL/pseudo-label-based training can suffer from error propagation.

In parallel, advances in graph reasoning—particularly dynamic graph structure learning and inference [15], [16]—provide powerful tools for relational modeling. Motivated by these developments, we propose a staged pipeline comprising spatial feature extraction, temporal feature extraction, and high-level reasoning to better decompose video content for accurate and comprehensive anomaly recognition. Specifically, we introduce S-TRGR, an end-to-end Spatio-Temporal Relation Graph Reasoning framework that targets mid-level semantic modeling by combining ResNet-based spatial convolutions for local feature extraction, a Transformer module to capture long-range temporal dependencies, and a Graph Convolutional Network to reason over spatio-temporal interactions. This unified design integrates local and global cues to improve abnormal behavior recognition in surveillance videos. The main contributions of this paper are summarized as follows:

- **Mid-level Semantic Modeling:** To the best of our knowledge, S-TRGR is the first framework to explicitly leverage mid-level video semantics to bridge visual features and behavior categories in abnormal event recognition.
- **Hierarchical S-TRGR Architecture:** We introduce a novel hierarchical framework that synergistically combines CNNs, Transformers, and GCNs to preserve fine-grained spatial details while enabling complex spatio-temporal reasoning.
- **State-of-the-art Performance:** Extensive evaluations on standard benchmarks (UCF-Crime [17] and XD-Violence [18]) validate the efficacy of S-TRGR, showing significant gains over existing attention-based and convolutional models.

II. METHODOLOGY

A. Methodological Overview

Video abnormal behavior recognition is a key task in intelligent surveillance, aiming to identify events that deviate from normal patterns. Existing methods remain limited: detection-based methods and 3D CNNs often fail to capture long-range spatio-temporal dependencies, while graph-based

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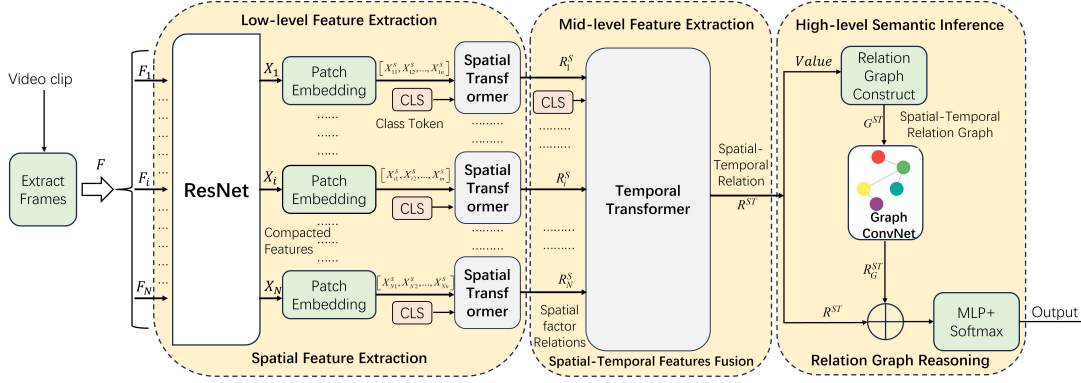


Fig. 1: S-TRGR framework overview, comprising the static feature extraction, interaction graph construction, and graph reasoning modules.

approaches depend heavily on object detection quality. To address these issues, we propose S-TRGR (Spatio-Temporal Relation Graph Reasoning), a unified framework that combines multilevel feature extraction and graph reasoning to model both local details and global dynamics. Given an input frame sequence $F = [F_1, F_2, \dots, F_N]$, our framework consists of three cascaded modules: **Static Feature Extraction:** A 2D convolutional backbone combining ResNet and spatial Transformer architectures extracts robust low-level visual features from individual frames. **Interaction Graph Construction:** Extracted features are spatially partitioned and encoded into sequence representations. A dual-transformer structure captures intra-frame spatial relations and inter-frame temporal dependencies, producing mid-level spatiotemporal features. These features are then projected into Query, Key and Value matrices to construct a unified spatiotemporal interaction graph. **Graph Reasoning and Classification:** A GCN is applied to the constructed graph for relational reasoning. The resulting embeddings are fused with the mid-level features to obtain high-level semantic representations, which are then fed to an MLP followed by a Softmax layer for classification.

B. Static Feature Extraction

While Transformers are effective at capturing global dependencies, they provide limited spatial inductive bias when processing raw pixels; conversely, CNNs naturally encode hierarchical local patterns. To leverage both properties, we adopt a CNN–Transformer cascade: a ResNet-50 backbone first extracts fine-grained spatial features, which serve as “visual tokens” enriched with structural priors. These tokens are subsequently fed into a Spatial Transformer to model global semantic correlations among different intra-frame regions. This design enhances the spatial representation capability of static features while preserving critical local details, thereby establishing a solid foundation for subsequent temporal interaction modeling.

C. Spatio-temporal Interaction Graph Construction

A graph $G = (Z, E)$ consists of a set of nodes $Z = \{z_1, z_2, \dots, z_P\}$, where each node is associated with a C -dimensional feature and can be stacked as $Z \in \mathbb{R}^{P \times C}$, and a set of edges E specifying pairwise connections. The graph

topology is represented by an adjacency matrix $A \in \mathbb{R}^{P \times P}$ induced by E , where $A_{i,j} = 1$ if $(z_i, z_j) \in E$ and 0 otherwise. Based on this formulation, we construct spatio-temporal interaction graphs from videos in two steps: (1) extract interaction elements with a spatio-temporal Transformer; and (2) form relational graphs from the resulting interaction patterns.

Spatio-temporal Interaction Relations Extraction Motivated by hierarchical cognition and ViViT [12], we employ a dual-branch Transformer to extract mid-level semantic features as graph nodes. Given CNN feature maps, we uniformly partition them along the spatial dimensions and add learnable positional embeddings. The architecture consists of a spatial Transformer for modeling intra-frame relations and a temporal Transformer for capturing inter-frame dependencies. A CLS token is used at each stage to aggregate token features into compact representations: the spatial Transformer generates frame-level embeddings, and the temporal Transformer further produces a global representation for cross-frame dynamics. To align the feature dimensions between the two stages, we apply average pooling to the spatial features. Overall, the ResNet–Transformer stack preserves both local details and global context with limited information loss. The extraction module is illustrated in Fig. 2a.

To model spatio-temporal interaction features, we employ a Pre-Layer Normalization (Pre-LN) Transformer with l stacked blocks (Fig. 2b). Each block comprises a self-attention sub-layer and a feed-forward sub-layer; both are preceded by layer normalization and followed by residual connections. This Pre-LN formulation improves optimization stability and typically converges faster than the Post-LN variant when learning complex temporal behavior patterns.

The Transformer adopts multi-head attention, which runs multiple scaled dot-product attention operations in parallel to capture relationships in different representation subspaces (Fig. 2c,d). Given an input sequence, the Query (Q), Key (K), and Value (V) matrices are computed from the same input features through separate learned linear projections and are then reshaped into multiple heads to obtain head-specific representations. The projected features are reshaped into n heads: $Q = [q_1, q_2, \dots, q_n]^T$, $K = [k_1, k_2, \dots, k_n]^T$, $V = [v_1, v_2, \dots, v_n]^T$. For each head i , attention weights are computed as $\text{Softmax}(q_i K^T / \sqrt{d})$, and the corresponding

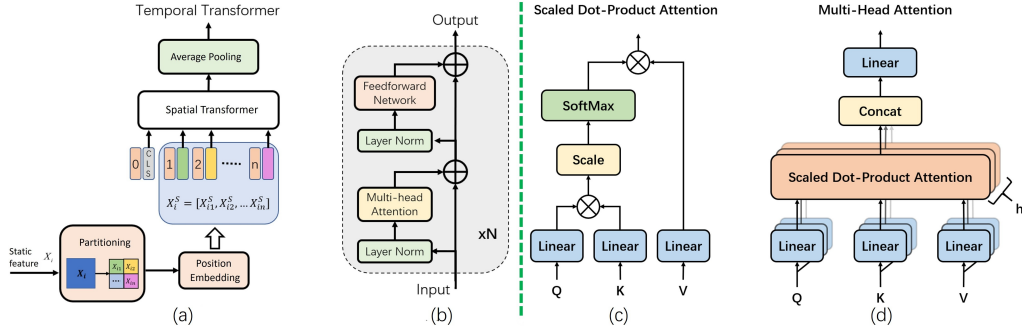


Fig. 2: Module details: (a) spatial branch; (b) Pre-LN Transformer block; (c) scaled dot-product attention; (d) multi-head attention.

output is obtained by weighting values with these scores, i.e., $\text{Softmax}(q_i K^T / \sqrt{d})V$. Our Pre-LN method applies layer normalization to inputs before computation of attention weights. With keys K projected into d_k -dimensional space, the scaled dot-product attention is formulated as:

$$\text{Attn}(Q, K, V) = \text{softmax}\left(\frac{QK^T}{\sqrt{d_k}}\right)V. \quad (1)$$

Multi-head attention extends this by projecting inputs into h distinct subspaces. Queries, keys, and values are linearly transformed by learned matrices and processed in parallel. Formally, the i -th attention head is computed as:

$$\text{head}_i = \text{Attn}\left(QW_i^Q, KW_i^K, VW_i^V\right), \quad (2)$$

where the outputs from all heads are subsequently concatenated and projected to produce the final representation. In this equation, $W_i^Q, W_i^K, W_i^V \in \mathbb{R}^{d_m \times d_k}$ denote the learnable projection matrices for Q , K , and V , respectively, where d_m is the model dimension and d_k signifies the dimensionality of each attention head. The outputs of all attention heads are concatenated and projected to form the final representation:

$$\text{MultiHead}(Q, K, V) = \text{Concat}(\text{head}_1, \dots, \text{head}_h). \quad (3)$$

Each Transformer block preserves a fixed feature dimensionality through a standard sequence of operations: multi-head self-attention followed by a residual connection, and then a two-layer feed-forward network with another residual connection. This design maintains dimensional consistency across layers while facilitating stable gradient propagation. **Spatio-temporal Interaction Graph Construction** We derive a spatio-temporal interaction graph $G = (Z, E)$ to support higher-order relational reasoning. In this framework, the Transformer module serves as a context-aware feature encoder: its internal self-attention mechanism enables each spatial/temporal token to aggregate global dependencies. The resulting attention-weighted Value projections are extracted as node embeddings, $Z = \phi_Z(R^{ST})$, which encode context-rich spatio-temporal information.

The spatio-temporal module outputs $R^{ST} \in \mathbb{R}^{B \times N' \times C}$, where B is batch size, $N' = N + 1$ includes N frames plus a temporal CLS token, and C is the feature dimension. For each video, we build a graph $G = (Z, E)$ with nodes:

$$Z = \text{Value} = \phi_Z(R^{ST}), \quad (4)$$

where ϕ_Z is the Value projection, yielding $Z \in \mathbb{R}^{N' \times C}$. We then compute an adjacency matrix A whose edge weights are the inverse Euclidean distances between node embeddings, following [15], thereby encoding semantic proximity among spatio-temporal elements.

Distinct from the implicit attention maps within the Transformer, the explicit relational topology for the subsequent GCN is constructed by measuring the semantic proximity between these node embeddings. Specifically, we compute the edge weight between nodes z_i and z_j as the inverse Euclidean distance in a learnable projected feature space:

$$f(z_i, z_j) = \frac{1}{\|\phi_A(z_i) - \phi_A(z_j)\|_2}, \quad (5)$$

where $\phi_A(z) = W_A \cdot z$ represents a learnable projection that optimizes the modeling of semantic relationships. The adjacency matrix is subsequently normalized via Softmax (Eq. 6) to ensure stable message passing during graph convolution.

$$A_{i,j} = \frac{\exp(f(z_i, z_j))}{\sum_{k=1}^N \exp(f(z_i, z_k))}, \quad \forall i, j \in \{1, 2, \dots, N\}. \quad (6)$$

We process the spatio-temporal graph $G^{ST} = (Z, A)$ with GCNs, using a symmetric adjacency ($A_{i,j} = A_{j,i}$) without self-loops ($A_{i,i} = 0$). Normalization stabilizes gradients and yields uniform message passing. Multi-hop graph convolutions produce refined features R_G^{ST} that capture both local temporal dependencies and global semantic interactions.

D. Graph Reasoning

We apply GCNs on the spatio-temporal interaction graph $G^{ST} = (Z, A)$ to model relational dependencies and obtain high-level semantic representations. The graph convolution is generally defined as $H^{l+1} = \sigma(AH^l W^l)$, where W^l denotes the learnable parameters. In practice, we adopt the symmetric normalized form

$$H^{l+1} = \sigma(\tilde{D}^{-1/2} \hat{A} \tilde{D}^{-1/2} H^l W^l), \quad (7)$$

where $\hat{A} = A + I_n$ denotes the adjacency matrix with self-loops. The diagonal degree matrix $\tilde{D}$ is constructed as follows:

$$\tilde{D}_{i,j} = \begin{cases} \sum_j \hat{A}_{i,j} & \text{if } i = j, \\ 0 & \text{otherwise.} \end{cases} \quad (8)$$

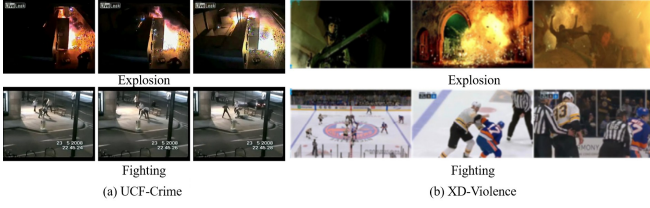


Fig. 3: Representative video frame samples from the UCF-Crime and XD-Violence datasets.

This normalization ensures that the scale of feature embeddings remains consistent during message passing. Specifically, the elements of the symmetric normalized Laplacian $L^{sym} = \tilde{D}^{-1/2} \hat{A} \tilde{D}^{-1/2}$ are computed to aggregate neighbor information while preserving the node’s individual identity:

$$L_{i,j}^{sym} = \begin{cases} 1 & \text{if } i = j \text{ and } \tilde{D}_{i,i} \neq 0, \\ -\frac{\hat{A}_{i,j}}{\sqrt{\tilde{D}_{i,i} \tilde{D}_{j,j}}} & \text{if } i \neq j \text{ and } (z_i, z_j) \in E, \\ 0 & \text{otherwise.} \end{cases} \quad (9)$$

After graph convolution, we fuse the GCN output with the spatio-temporal features via element-wise addition. The fused representation is average-pooled and passed through an MLP with a Softmax layer for final classification. The model is trained using cross-entropy loss:

$$\mathcal{L}_{CE}(\mathbf{y}, \hat{\mathbf{y}}) = -\frac{1}{M} \sum_{i=1}^M \sum_{j=1}^C y_{i,j} \log(\hat{y}_{i,j}), \quad (10)$$

where $y_{i,j}$ denotes the ground-truth label and $\hat{y}_{i,j}$ denotes the predicted probability. This fusion enables the model to jointly leverage local spatio-temporal cues and global graph-structured relations, resulting in more robust abnormal behavior classification.

III. EXPERIMENTS

A. Experimental Setup

We conduct experiments on two benchmarks, UCF-Crime [17] and XD-Violence [18]. UCF-Crime is particularly challenging: it consists of untrimmed surveillance videos with an average duration of approximately 4 minutes and covers 13 abnormal event categories. This setting contrasts with conventional action-recognition datasets such as UCF-101 [19], which comprise short (10–20s) clips of isolated actions. UCF-Crime contains 1,900 videos (Fig. 3a), requiring robust modeling of long-range temporal patterns in complex real-world scenes. For XD-Violence, the largest publicly available violence-detection dataset, we use 3,954 training videos and 800 test videos (500 violent and 300 non-violent; see Fig. 3b) collected from movies and YouTube and covering diverse scenarios. Fair comparisons with state-of-the-art methods on both benchmarks are conducted to comprehensively evaluate the proposed method.

We adopt a 5:1 train/test split. For UCF-Crime, we randomly sample 64-frame clips during training and use three uniformly spaced segments for testing. For XD-Violence, videos are standardized to 24 FPS and processed with a 16-frame sliding window, together with five-crop augmentation.

Confusion matrix (%)	P1	P2	P3	P4	P5	P6	P7	P8	P9	P10	P11	P12	P13
R1	100	5.6	0	5.6	12.1	0	11.1	11.1	22.2	0	5.6	5.6	5.6
R2	0	22.2	5.6	0	11.1	11.1	0	16.7	27.8	0	5.6	0	0
R3	1.9	1.9	51.8	21.9	3.7	5.6	0	5.6	0	3.7	3.7	1.9	0
R4	0	33.0	3.7	22.2	11.1	5.6	0	5.6	22.2	0	0	11.1	5.6
R5	0	0	0	2.8	38.9	0	2.8	16.7	11.1	0	11.1	13.9	2.8
R6	0	0	11.1	0	22.2	22.2	0	5.6	16.7	0	0	16.7	5.6
R7	0	5.6	0	11.1	11.1	0	22.2	22.2	22.2	0	0	5.6	0
R8	0	0	9.5	0	9.5	3.7	0	45.1	7.4	0	0	20.4	3.7
R9	5.6	3.7	0	1.9	16.7	0	0	11.1	33.3	0	5.6	22.2	0
R10	16.7	5.6	0	0	0	0	0	16.7	11.1	27.8	0	16.7	5.6
R11	0	3.7	5.6	18.9	16.7	11.1	0	0	22.2	0	16.7	5.6	0
R12	2.8	2.8	8.3	2.8	0	0	5.6	5.6	22.2	0	2.8	47.2	0
R13	0	0	5.6	5.6	0	0	0	16.7	11.1	0	0	27.8	33.3

(a) X3D-XL method

Confusion matrix (%)	P1	P2	P3	P4	P5	P6	P7	P8	P9	P10	P11	P12	P13
R1	100	9.1	0	0	9.1	0	9.1	9.1	0	0	0	0	0
R2	0	45.4	9.1	0	9.1	0	0	27.3	9.1	0	0	0	0
R3	0	0	45.4	0	18.2	9.1	0	0	0	0	0	0	27.3
R4	9.1	18.2	0	18.2	0	9.1	9.1	9.1	0	0	18.2	0	0
R5	10.0	0	0	4.8	36.1	0	9.5	0	9.5	0	0	14.3	4.8
R6	0	0	0	0	0	0	36.1	0	27.3	9.1	0	0	9.1
R7	0	18.2	0	18.1	9.1	0	18.2	9.1	9.1	0	9.1	0	0
R8	3.2	3.2	3.2	3.2	6.4	12.9	0	6.4	6.4	0	3.2	0	3.2
R9	12.9	3.2	0	0	6.4	3.2	0	12.9	6.4	6.4	3.2	3.2	0
R10	9.1	9.1	0	0	18.2	0	0	9.1	9.1	18.2	0	9.1	18.2
R11	0	0	9.1	0	0	0	0	0	18.2	0	6.4	9.1	0
R12	4.8	0	4.8	0	4.8	0	4.8	0	19.0	4.8	0	9.5	92.4
R13	9.1	9.1	9.1	9.1	0	0	0	9.1	9.1	0	0	9.1	36.3

(b) S-TRGR method

Fig. 4: Confusion matrices: X3D-XL vs. S-TRGR on UCF-Crime. The horizontal axis (P) represents predicted labels, while the vertical axis (R) represents real/ground truth labels. In the confusion matrix, cell color intensity corresponds to value, with darker shades indicating higher counts. The 13 class labels (1–13) are: Abuse, Arrest, Arson, Assault, Burglary, Explosion, Fighting, Road Accidents, Robbery, Shooting, Shoplifting, Stealing, and Vandalism. In (a), red text highlights the three classes most frequently misclassified as Robbery (class 9), along with their misclassification percentages.

TABLE I: Classification accuracy on UCF-Crime compared with state-of-the-art methods.

Method	Top-1(%)	Top-5(%)
C3D [6]	23.0	64.9
I3D [7]	27.2	72.5
R(2+1)D [8]	27.6	70.3
STRG [20]	29.8	78.6
SlowFast [9]	32.3	81.2
MAF [21]	31.0	—
X3D-XL [10]	34.2	80.6
WJADC [5]	34.1	—
ABRF [1]	37.9	—
S-TRGR	42.9	82.6

All models are trained on four NVIDIA Titan XP GPUs using SGD with momentum 0.9, an initial learning rate of 0.001 decayed by a factor of 10, and class-balanced (weighted) sampling to mitigate class imbalance.

B. Results on UCF-Crime Dataset

We evaluate our method on the UCF-Crime dataset by benchmarking it against state-of-the-art approaches. These include 3D convolutional models (C3D [6], I3D [7], R(2+1)D [8], SlowFast [9], and X3D-XL [10]), Siamese 3D CNN [2], the graph-based model STRG [20], and attention-driven methods (MAF [21], WJADC [5], and ABRF [1]). In addition to conventional CNN-based methods, we also compare S-TRGR with recent advanced approaches, including VadCLIP [3] and Weakly Supervised Audio-Visual Violence Detection (WSAV) [4]. Performance is evaluated using Top-1 accuracy, Top-5 accuracy, and mAP. Results for MAF, WJADC, and ABRF, VadCLIP, WSAV are directly obtained from related publications, while our experiments for STRG, C3D, I3D, R(2+1)D, SlowFast, and X3D are conducted under identical conditions. As shown in Table I, the S-TRGR approach outperforms current state-of-the-art methods on the UCF-Crime dataset by achieving a 5.0% improvement in Top-1 and a 1.4% gain in Top-5 accuracy. This performance boost confirms S-TRGR’s capability in capturing complex temporal dynamics and semantic representations in long-duration video sequences for abnormal behavior recognition.

We further report mean average precision (mAP), computed as the mean of per-class AP. We include the multi-

TABLE II: mAP comparison on UCF-Crime against state-of-the-art methods.

Method	mAP(%)
R(2+1)D [8]	21.5
CNN-RNN [23]	22.9
STRG [20]	26.3
SlowFast [9]	28.6
X3D-XL [10]	31.7
MLIR [22]	36.1
Siamese 3D CNN [2]	38.4
TimeSformer [11]	31.5
ViViT [12]	34.1
WSAV [4]	30.5
VadCLIP [3]	37.0
S-TRGR	42.9

level interaction reasoning method (MLIR) [22] and the CNN-RNN hybrid [23] for comparison, as both follow a similar paradigm of global information extraction followed by reasoning for video understanding. Table II summarizes the results on UCF-Crime, where the proposed method achieves the highest mAP among all compared methods.

Fig. 4 compares confusion matrices of X3D-XL [10] (a) and the S-TRGR model (b) on UCF-Crime. Our model achieves higher classification accuracy (max: 63.6%, min: 18.2%, five categories >50%) than X3D-XL (max: 51.4%, min: 16.7%, only one category >50%). S-TRGR shows >10% improvement on complex, long-duration behaviors (Abuse [R1], Arrest [R2], Explosion [R6], Shoplifting [R11]), while X3D-XL performs marginally better on short-duration, scene-specific actions (Arson [R3], Assault [R4], Fighting [R7], Shooting [R10]). This demonstrates S-TRGR’s advantage in handling ambiguous, dynamic scenes compared to X3D-XL’s reliance on spatial cues in well-defined events. Moreover, X3D-XL exhibits a stronger bias toward classes with more training samples (e.g., Road Accidents [R8], Robbery [R9], Stealing [R12]), whereas S-TRGR produces more balanced predictions, with clearer diagonal dominance and fewer cross-class confusions (e.g., reduced confusion among Abuse [R1], Arrest [R2], and Robbery [R9]). These results support that enhanced long-range dependency modeling and dynamic attention in S-TRGR lead to improved classification performance.

C. Results on XD-Violence Dataset

We evaluate S-TRGR on XD-Violence to assess its generalization across diverse content. We compare against representative baselines, including reconstruction-based unsupervised learning [24], weakly supervised MIL with video-level labels [17], causal temporal modeling with feature discrimination [25], contrastive attention for handling class imbalance [13], Transformer-based multi-sequence learning [14], multimodal audio-visual fusion with relational modeling (WSAV) [4], and approaches that adapt pre-trained vision-language models [26]. Covering reconstruction, weak supervision/ranking, temporal modeling, multimodal fusion, and attention mechanisms, this set of baselines provides a comprehensive evaluation. The results are reported in Table III. S-TRGR achieves 78.7% mAP on XD-Violence,

TABLE III: mAP comparison on XD-Violence against state-of-the-art methods.

Method	mAP(%)
Hasan et al. [24]	30.8
Sultani et al. [17]	73.2
Chang et al. [13]	76.9
Ju et al. [26]	76.6
Wu and Liu [25]	75.9
Wu et al. [4] (WSAV)	78.6
Li et al. [14]	78.6
S-TRGR	78.7

substantially outperforming earlier methods such as Hasan et al. [24] (30.8%) and Sultani et al. [17] (73.2%), as well as more recent approaches including Chang et al. [13] (76.9%), Ju et al. [26] (76.6%), and Wu and Liu [25] (75.9%). It also slightly surpasses the strongest competitors, WSAV [4] (78.6%) and Li et al. [14] (78.6%), yielding the best overall performance and supporting the effectiveness of our spatio-temporal reasoning design.

For statistical robustness, we report the mAP of 78.7% as the average over five independent runs. Although the margin over WSAV is small, the comparison remains meaningful because S-TRGR relies only on visual input, whereas WSAV uses both audio and visual modalities. Surpassing a multi-modal state-of-the-art method with visual data alone suggests that the proposed spatio-temporal reasoning effectively captures rich relational cues from video frames.

Together with the UCF-Crime results, these findings show that S-TRGR generalizes well across datasets with different characteristics. The consistent gains on both benchmarks highlight the effectiveness of graph-based spatio-temporal modeling for violence detection in unconstrained videos.

D. Ablation Study

We perform ablation studies on three components—spatio-temporal Transformers, backbone variants, and GCN modules—to evaluate their roles in mitigating attention overfitting and preserving robust representations.

Use spatial and temporal Transformers to extract mid-level features: Our method models mid-level spatio-temporal interactions for graph construction and semantic reasoning. Specifically, we employ a two-stage Transformer, in which a spatial Transformer captures intra-frame relations and a temporal Transformer integrates cross-frame information to model long-range dependencies beyond CNNs. As shown in Table IV (Col. 4), both modules outperform average pooling, yielding gains of +7.8% (spatial) and +13.2% (temporal), and +19.3% when combined. The smaller improvement from the spatial module likely stems from redundancy in ResNet features, whereas the temporal module contributes most by modeling complex temporal dynamics. On XD-Violence (Col. 5), accuracy increases from 63.1% to 75.6% with the spatial module and to 76.8% with the temporal module, reaching 78.7% when both are enabled.

Use backbones to extract static features: Extensive ablations (Table V, Col. 3) indicate that incorporating a backbone substantially improves static feature extraction and stabi-

TABLE IV: Ablation results for the spatio-temporal modules and their combinations.

	Spatial-T	Temporal-T	UCF-Crime	XD-Violence
1			23.6	63.1
2	✓		31.4	75.6
3		✓	36.8	76.8
4	✓	✓	42.9	78.7

TABLE V: Ablation results for different backbone choices.

	Backbone	UCF-Crime	XD-Violence
1	None	16.1	9.1
2	CNN	24.3	38.7
3	ResNet18	37.9	66.3
4	ResNet34	39.4	71.5
5	ResNet50	42.9	78.7
6	ResNet101	42.7	77.1

lizes Transformer training, particularly under limited training data. We compare a no-backbone setting, a lightweight CNN, and ImageNet-pretrained ResNets (18/34/50/101). Pre-training proves crucial for providing strong spatial features, which are further leveraged by the temporal Transformer. Performance improves consistently from ResNet18 to ResNet34/50, while ResNet101 yields only marginal additional gains. Considering the accuracy–cost trade-off, we adopt ResNet50. A similar saturation trend is observed on XD-Violence, consistent with the backbone-scaling behavior.

S-TRGR has a total model size of 974 MB, mainly due to the use of ResNet-50, dual Transformer modules, and a GCN for spatio-temporal reasoning in unconstrained surveillance. Despite its multi-stage architecture, patch embedding and the CLS token help preserve compact and informative representations.

Use GCN for high-level semantic reasoning: Our spatio-temporal interaction graph captures mid-level semantics, and the graph reasoning module infers higher-level relations through structured feature fusion. On UCF-Crime (Table VI), adding this module improves performance by 2.6% without static features and by 4.5% with them, suggesting enhanced behavior interpretation. On XD-Violence, incorporating the GCN without a backbone increases accuracy from 39.1% to 50.8%; with a ResNet-50 backbone, the gain is smaller (77.9%→78.7%), indicating that strong backbones already capture some of the structural cues modeled by our GCN.

IV. CONCLUSIONS

In this paper, we present S-TRGR, a framework that leverages mid-level semantics to improve abnormal behavior recognition in challenging surveillance videos. S-TRGR integrates a hierarchical pipeline of ResNet, spatio-temporal Transformers, and GCN-based reasoning to capture local spatial cues and long-range temporal dependencies via graph-structured interactions. Extensive experiments on two large-scale benchmarks demonstrate that S-TRGR consistently improves recognition performance. In future work, we will incorporate multimodal cues to further enhance event understanding in unconstrained environments.

TABLE VI: Ablation results for the GCN-based reasoning module.

	Backbone	S-T	GCN	UCF-Crime	XD-Violence
1		✓		13.5	39.1
2		✓	✓	16.1	50.8
3	✓	✓		38.4	77.9
4	✓	✓	✓	42.9	78.7

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